

Limits & variations of linear regression

Readings for today

- Chapter 3: Linear regression. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning: with applications in R (Vol. 6). New York: Springer.

Topics

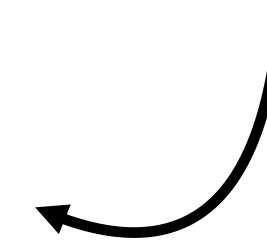
1. Interpretation constraints
2. Pitfalls of linear regression

Interpretation constraints

Goals of learning f

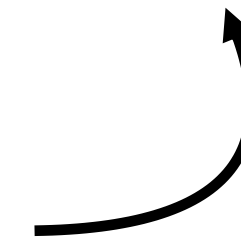
1. Inference: understand how changes in X associate with changes in Y .

Intuition



2. Prediction: predict a future observation in Y from X .

Control



Example: Inference

Q: What factors associate with house value?

Model:

$$Y_{\$} = \beta_0 + \beta_{crime}X_{crime} + \beta_{age}X_{age} + \beta_{tax}X_{tax} + \beta_{s:t}X_{s:t}$$

Variables:

- $Y_{\$}$: median house value
- X_{crime} : local crime rate
- X_{age} : # houses over 80 years old
- X_{tax} : local property tax rate
- $X_{s:t}$: school student-teacher ratio

	$\hat{\beta}_i$	$SE(\hat{\beta}_i)$	t	p	sig
Intercept	36.46	5.10	7.14	3.28E-12	***
Crime	-1.08	0.33	-3.29	0.002	**
Age	0.01	0.01	0.052	0.85	
Tax	-0.12	0.01	-3.28	0.001	**
S:T ratio	-0.95	0.13	-7.28	0.0005	***

Example: Prediction

Q: What factors predict house value?

Model:

$$Y_{\$} = \hat{\beta}_0 + \hat{\beta}_{crime}X_{crime} + \hat{\beta}_{age}X_{age} + \hat{\beta}_{tax}X_{tax} + \hat{\beta}_{s:t}X_{s:t}$$

Steps:

1. Fit $\hat{f}_{train}(X_{train})$
2. Predict $\hat{Y}_{test}$ using $\hat{f}_{train}(X_{test})$
3. Evaluate test error:

$$\sum_{i=1}^n (y_i^{test} - \hat{y}_i^{test})^2$$

$$Y_{train}, X_{train} \begin{pmatrix} y_i \\ \vdots \\ y_m \\ y_{m+1} \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} & x_{1,5} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{m,1} & x_{m,2} & x_{m,3} & x_{m,4} & x_{m,5} \\ x_{m+1,1} & x_{m+1,2} & x_{m+1,3} & x_{m+1,4} & x_{m+1,5} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_{n,1} & x_{n,2} & x_{n,3} & x_{n,4} & x_{n,5} \end{pmatrix} \quad \text{Training set}$$

Categorical predictors

Q: Are there gender differences in reading comprehension?

Variables:

- Y_{read} : content recall score
- X_{gender} : gender (men, women)

Model:

$$Y_{read} = \hat{\beta}_0 + \hat{\beta}_1 X_{gender}$$

Case 1: Dummy coding

$$X_{gender} \begin{cases} 0, men \\ 1, women \end{cases}$$

$$men = \hat{\beta}_0$$

$$women = \hat{\beta}_1$$

Case 2: Effect coding

$$X_{gender} \begin{cases} -1, men \\ 1, women \end{cases}$$

$$men = \hat{\beta}_0 - \hat{\beta}_1$$

$$women = \hat{\beta}_0 + \hat{\beta}_1$$

How you specify your qualitative predictor variables determines how you interpret the effects

Categorical predictors (>2 levels)

Q: Are there gender differences in reading comprehension?

Variables:

- Y_{read} : content recall score
- X_{gender} : gender (men, women, nonbinary)

Dummy (binary) coding:

$$X_{women} \begin{cases} 0, otherwise \\ 1, women \end{cases}$$

$$X_{nb} \begin{cases} 0, otherwise \\ 1, nonbinary \end{cases}$$

Model:

$$Y_{read} = \hat{\beta}_0 + \hat{\beta}_1 X_{women} + \hat{\beta}_2 X_{nb}$$

$$\text{men} = \hat{\beta}_0$$

$$\text{women} = \hat{\beta}_0 + \hat{\beta}_1$$

$$\text{nonbinary} = \hat{\beta}_0 + \hat{\beta}_2$$

For a k-level categorical variable, you include **k-1 dummy variables**, and each observation activates exactly one level.

Categorical predictors (>2 levels)

Q: Are there gender differences in reading comprehension?

Variables:

- Y_{read} : content recall score
- X_{gender} : gender (men, women, nonbinary)

Effect coding

$$X_{women \text{ diff } men?} \begin{cases} -1, men \\ 1, women \\ 0, nonbinary \end{cases} \quad X_{NB \text{ diff } binary?} \begin{cases} -1, men \\ -1, women \\ 2 nonbinary \end{cases}$$

Model:

$$Y_{read} = \hat{\beta}_0 + \hat{\beta}_1 X_{women \text{ diff } men?} + \hat{\beta}_2 X_{NB \text{ diff } binary?}$$

$$\hat{\beta}_0 = \text{overall mean across all genders}$$

$$\hat{\beta}_1 = \frac{\mu_{women} - \mu_{men}}{2}$$

$$\hat{\beta}_2 = \mu_{nb} - \frac{\mu_{men} + \mu_{women}}{2}$$

Interactions

Q: Does gender influence the effect of age on reading comprehension?

Variables:

- Y_{read} : content recall score
- X_{gender} : gender (men, women)
- X_{age} : age (in years)

Model:

$$X_{gender} \begin{cases} 0, men \\ 1, women \end{cases}$$

$$Y_{read} = \hat{\beta}_0 + \hat{\beta}_1 X_{gender} + \hat{\beta}_2 X_{age} + \hat{\beta}_3 X_{gender} X_{age}$$

main effect of gender main effect of age gender by age interaction

Hierarchical Principle:

When including interaction terms, *always* include the main effect terms in the model.

Why do we need the Hierarchical Principle?

$$\begin{aligned} Y &= \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_1 X_2 \\ &= \hat{\beta}_0 + \underbrace{(\hat{\beta}_1 X_1 + \hat{\beta}_3 X_2)}_{\hat{\beta}_1^*} X_1 + \hat{\beta}_2 X_2 \\ &= \hat{\beta}_0 + \hat{\beta}_1^* X_1 + \hat{\beta}_2 X_2 \end{aligned}$$

- $\hat{\beta}_1^*$ depends on both X_1 and X_2 in order to describe the total relationship that X_1 has with Y .
- Excluding the main effect term $\hat{\beta}_1 X_1$ removes that portion of the variance explained by $\hat{\beta}_1^*$.

Pitfalls of linear regression

Structure of ordinary linear regression

Fundamental form:

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

Solution:

$$\hat{\beta}_0 = E[Y] + \hat{\beta}_1 E[X]$$

$$= \bar{y} + \sum_{j=1}^p \hat{\beta}_j \bar{x}_j$$

$$\hat{\beta}_j = \frac{\sum_{i=1}^n (x_{i,j} - \bar{x}_j)(y_i - \bar{y})}{\sum_{i=1}^n (x_{i,j} - \bar{x}_j)^2}$$

Necessary Assumptions to compute Ordinary Least Squares (OLS):

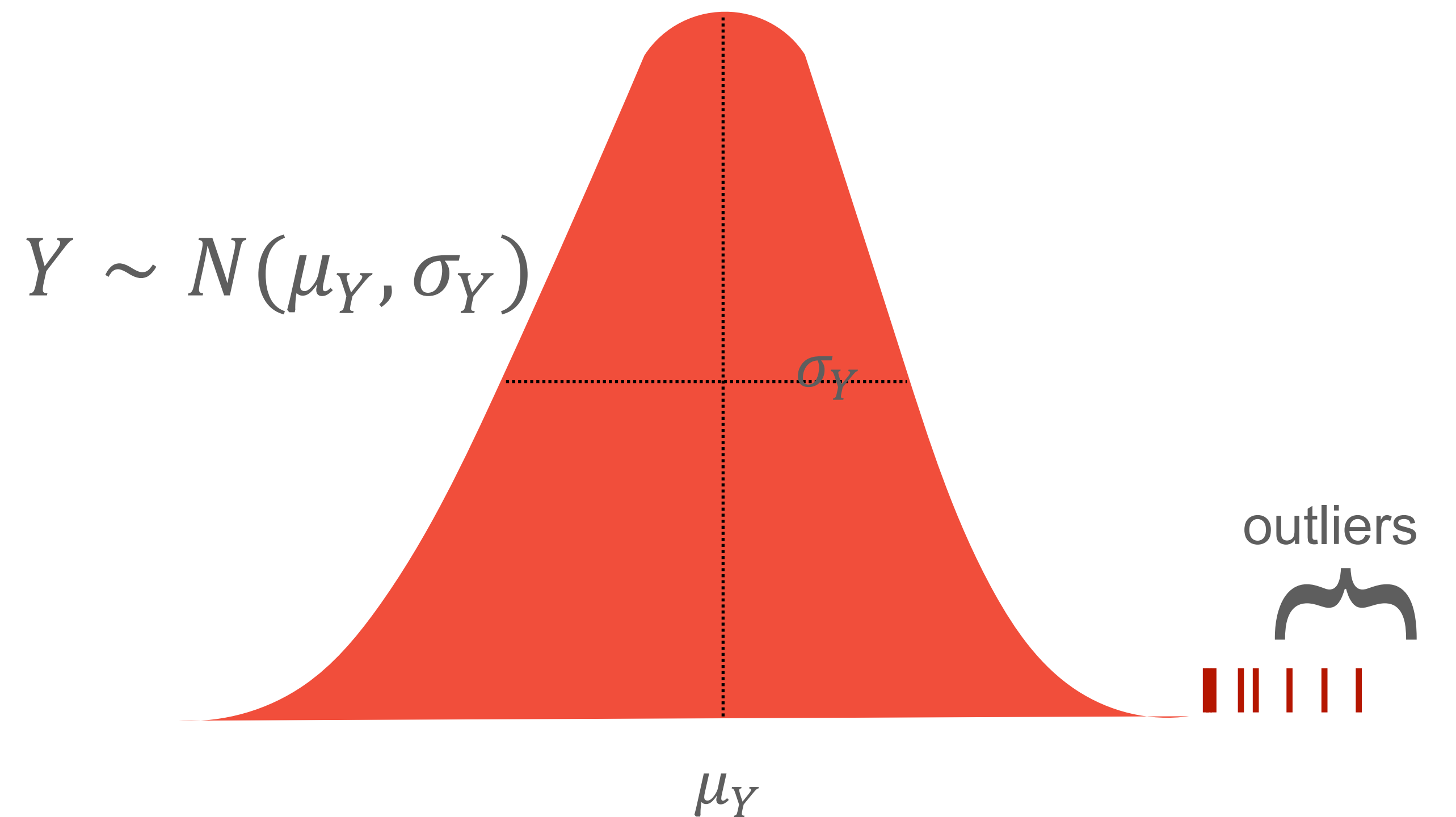
1. **Linear model specification:** predictors enter linearly in the parameters ($Y = X\beta + \epsilon$)
2. **Enough observations** ($n \geq p$): otherwise the columns of X cannot be linearly independent.
3. **No perfect multicollinearity:** No predictor is an exact linear combination of others.

+ Residuals of Y are normally distributed

+ For every value x_i there is only one y_i

Outliers

A subset of values that violate the assumed sampling distribution of Y .

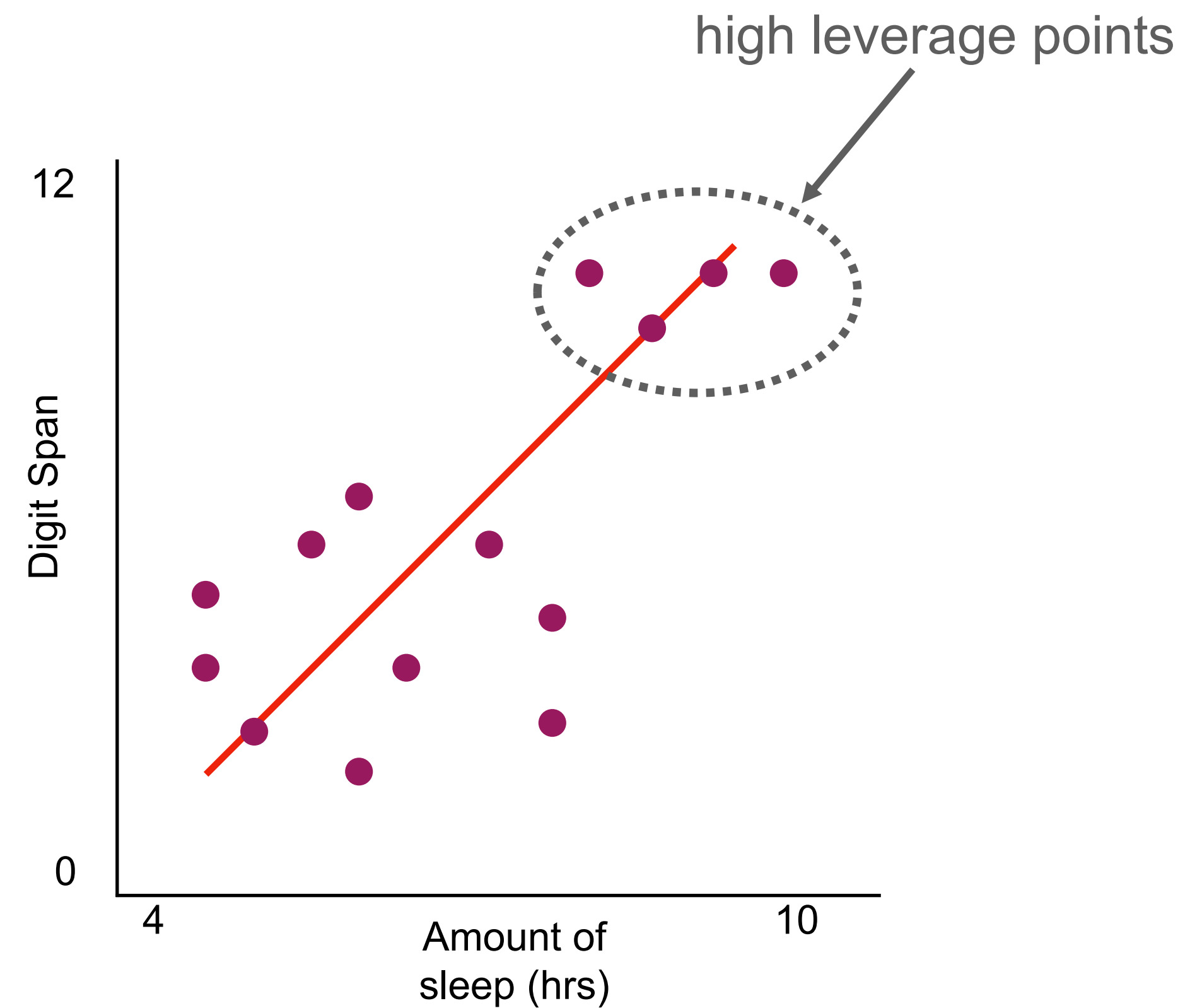
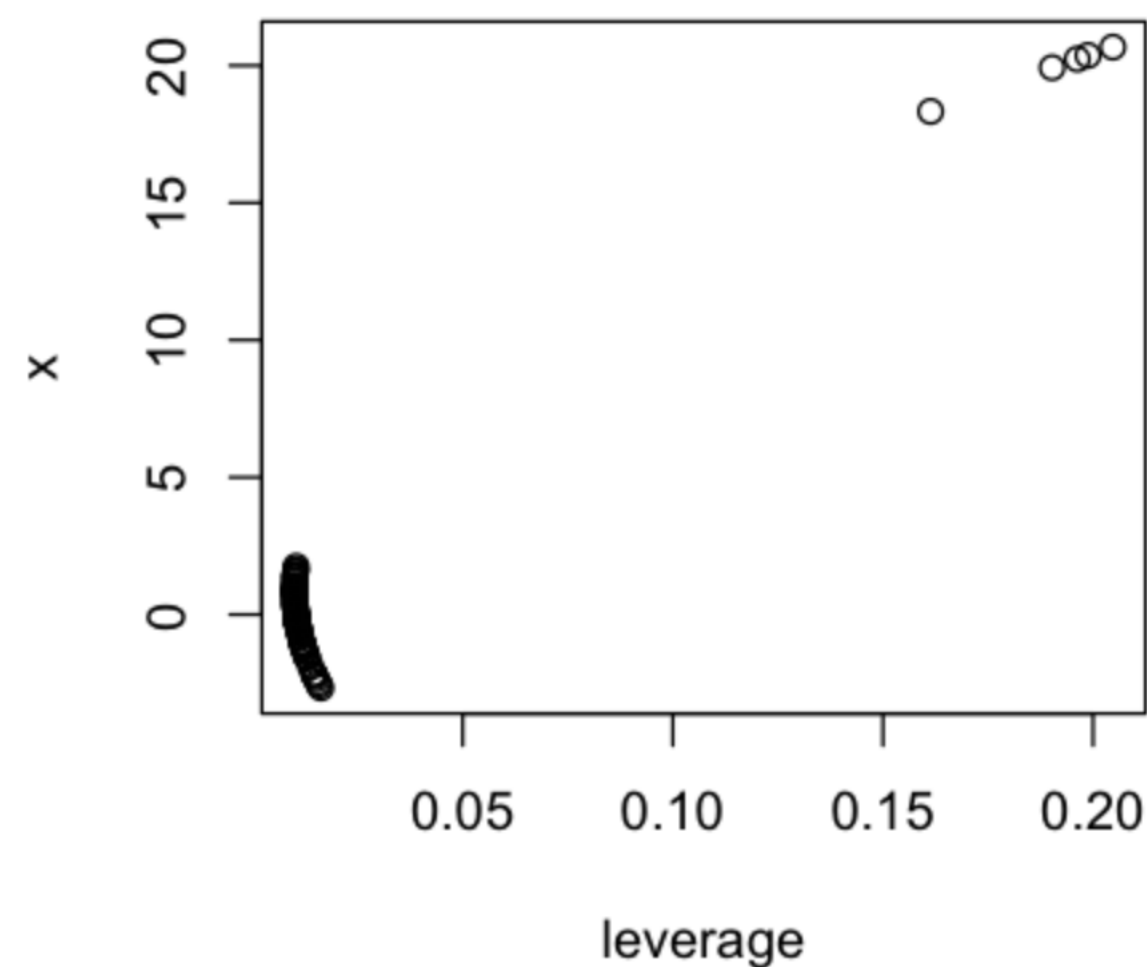


High leverage points

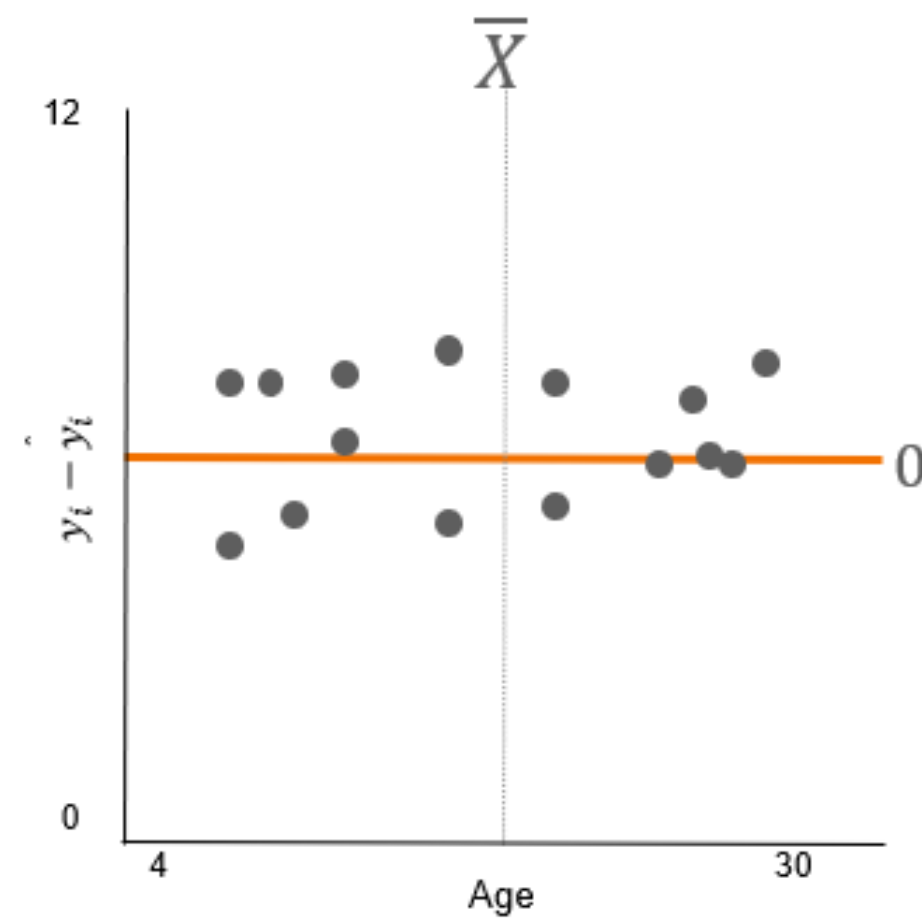
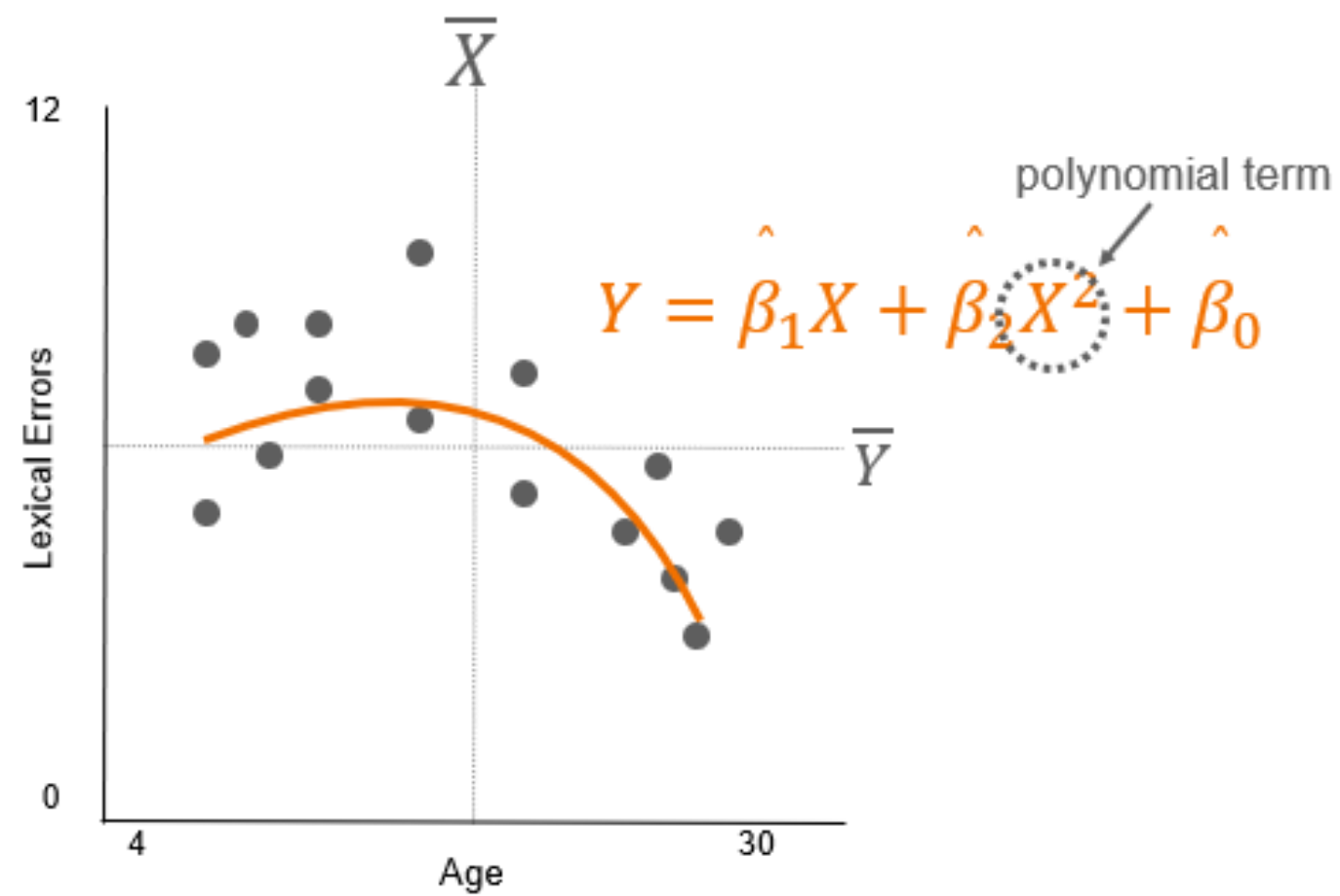
A subset of values in X that bias the outcome of $\hat{f}(X)$.

Leverage Statistic (h)

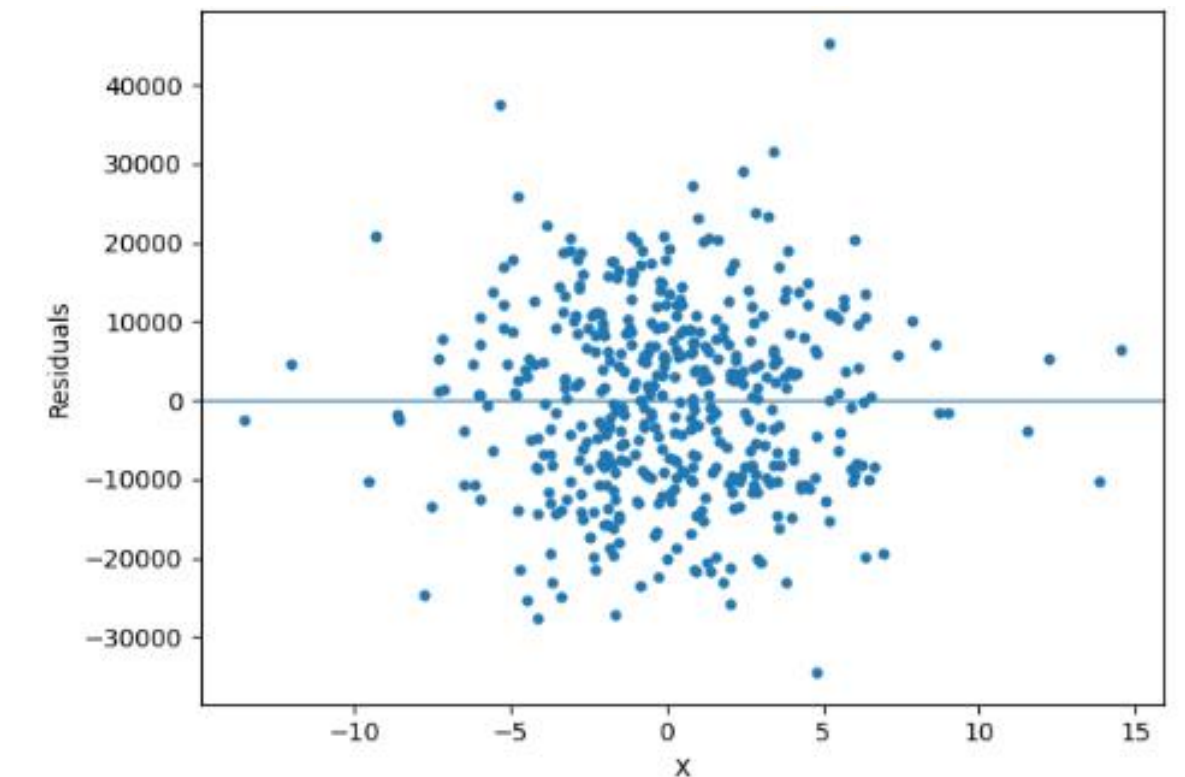
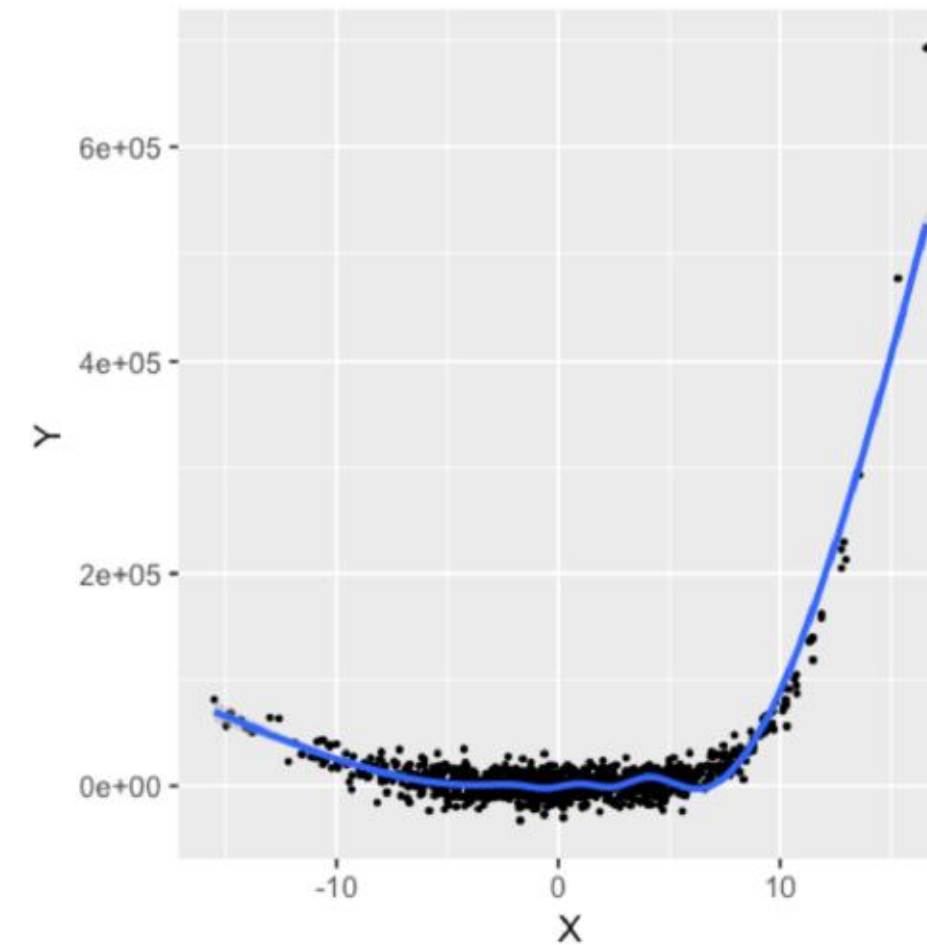
$$h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{i'=1}^n (x_{i'} - \bar{x})^2}$$



Leverage and across models



$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \beta_5 X^5$$



Collinearity

A correlation between two or more predictor variables in X .

Strong case: $Y = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2$

Set: $X_1 = X_2 \rightarrow \rho(X_1, X_2) = 1$

Then:
$$\begin{aligned} Y &= \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 \\ &= \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_1 \\ &= \hat{\beta}_0 + X_1 (\underbrace{\hat{\beta}_1 + \hat{\beta}_2}_{\tilde{\beta}_1}) \\ &= \hat{\beta}_0 + \tilde{\beta}_1 X_1 \end{aligned}$$

Variance Inflation Factor (VIF)

$$VIF = \frac{1}{1 - r_{X_j|X_{-j}}^2}$$

The closer to 1 that $r_{X_j|X_{-j}}^2$ gets, the more likely that the correlation is impacting $\hat{f}(X)$.

Take home message

- Interpreting linear regression models depends critically on how they are setup and how closely the data aligns to the assumptions of the fitting routine.